INCH-POUND
MIL-DTL-24643/22F
27 November 2023
SUPERSEDING
MIL-DTL-24643/22E
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 5,000 VOLTS, TYPE LS5KVTSGU (INCLUDING VARIATION LS5KVTSGUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LS5KVTSGU and LS5KVTSGUS (single overall shield).

Construction (Watertight with Circuit Integrity)

First - Copper conductor, coated or uncoated (see table I or table II for size).

Second - Thermoset insulation, extruded or taped. Minimum average wall thickness of 0.140 inch.

Third - An optional covering.

Fourth - The three conductors shall be cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm, well-rounded assembly. Standard identification code applied by method 1 3, or letter identification code applied by method 5; with one conductor each A (or black), B (or white), and C (or red). If identification method 1 is used, outer surface to be printed shall appear black.

Fifth - An optional binder.

Construction for Variant Type LS5KVTSGU

Sixth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.090 inch. See table I for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9).

Construction for Variant Type S5KVTSGUS

Sixth - Braided shield of tin-coated copper.

Seventh - An optional binder.

Eighth - Cross-linked polyolefin jacket. Absolute minimum average wall thickness 0.090 inch. See table II for cable overall diameter.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. LS5KVTSGU details.

Military part no. M24643/22	Type and size	Conductor size		Cable overall diameter max. (inches)	Conductor resistance (ohms)	Insulation resistance (megohms)	Accelerated service loading (amperes)
		Navy STD	AWG/MCM				
-01UN	LS5KVTSGU-100	--	0 (class D)	1.74	0.106	400	295
-02UN	LS5KVTSGU-150	--	000 (class D)	1.95	0.0674	350	400
-03UN	LS5KVTSGU-250	--	250 MCM (class C)	2.22	0.0453	300	535
-04UN	LS5KVTSGU-350	--	350 MCM (class D)	2.45	0.0321	265	665
-05UN	LS5KVTSGU-400	400 (127)	--	2.60	0.0273	250	750

TABLE II. LS5KVTSGUS details.

Military part no. M24643/22	Type and size	Conductor size		Cable overall diameter max. (inches)	Conductor resistance (ohms)	Insulation resistance (megohms)		Accelerated service loading (amperes)
		Navy STD	AWG/ MCM			Condr. to Condr.	Condr. to Shield	
-01UO	LS5KVTSGUS-100	--	0 (class D)	1.82	0.106	400	200	295
-02UO	LS5KVTSGUS-150	--	000 (class D)	2.03	0.0674	350	175	400
-03UO	LS5KVTSGUS-250	--	250 MCM (class C)	2.30	0.0453	300	150	535
-04UO	LS5KVTSGUS-350	--	350 MCM (class D)	2.53	0.0321	265	130	665
-05UO	LS5KVTSGUS-400	400 (127)	--	2.68	0.0273	250	125	750

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.) — See table I or table II

Voltage withstand (volts, root mean square, min.)

Conductor to conductor — 13,500

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Conductor to overall shield (type LS5KVTSGUS only)	5,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	See table I or table II
Conductor to overall shield (type LS5KVTSGUS only)	See table II
Conductor continuity	No failure
Shield continuity (type LS5KVTSGUS only)	No failure
Jacket flaws (type LS5KVTSGUS only)	No failure
Group A:	
Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Group B:	
Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation (extruded insulation only)	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C]) (see MIL-DTL-24643 for limits of drip allowance)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of print (insulation, method 1 only) (cycles, min.)	25

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Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Surface transfer impedance (type LS5KVTSGUS only)	
Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60

Group D:

Gas flame (circuit integrity) (1 hour)	No failure
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (-4±3.6 °F [-20±2 °C]) (type LS5KVTSGU only)	No damage
Gas flame (circuit integrity) (3 hours)	No failure
Accelerated service (sizes 100 and larger) (see table I or table II for load current)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25

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Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption (extruded insulation only)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS5KVTSGU all sizes	1,000
LS5KVTSGUS all sizes	1,000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-038)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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